

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

struct Node *head = NULL;

struct Node* createNode(int value) {
    struct Node* newNode = (struct Node*) malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->next = NULL;
    return newNode;
}

void createList(int n) {
    struct Node *newNode, *temp;
    int data, i;

    if (n <= 0) return;

    printf("Enter data for node 1: ");
    scanf("%d", &data);
    head = createNode(data);
    temp = head;

    for (i = 2; i <= n; i++) {
        printf("Enter data for node %d: ", i);
        scanf("%d", &data);
        newNode = createNode(data);
        temp->next = newNode;
        temp = newNode;
    }

    printf("List created successfully.\n");
}

void displayList() {
    struct Node *temp = head;
    if (head == NULL) {
        printf("List is empty.\n");
        return;
    }
    printf("Linked List: ");
    while (temp != NULL) {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}

void insertAtBeginning(int value) {
    struct Node* newNode = createNode(value);
    newNode->next = head;
    head = newNode;
    printf("Inserted %d at the beginning.\n", value);
}
```

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```
49     newNode->next = head;
50     head = newNode;
51     printf("Inserted %d at the beginning.\n", value);
52 }
53 void insertAtEnd(int value) {
54     struct Node *newNode = createNode(value), *temp;
55     if (head == NULL) {
56         head = newNode;
57     } else {
58         temp = head;
59         while (temp->next != NULL)
60             temp = temp->next;
61         temp->next = newNode;
62     }
63     printf("Inserted %d at the end.\n", value);
64 }
65 void insertAtPosition(int value, int pos) {
66     struct Node *newNode, *temp;
67     int i;
68
69     newNode = createNode(value);
70
71     if (pos == 1) {
72         newNode->next = head;
73         head = newNode;
74         printf("Inserted %d at position %d.\n", value, pos);
75         return;
76     }
77
78     temp = head;
79     for (i = 1; i < pos - 1 && temp != NULL; i++) {
80         temp = temp->next;
81     }
82
83     if (temp == NULL) {
84         printf("Invalid position!\n");
85     } else {
86         newNode->next = temp->next;
87         temp->next = newNode;
88         printf("Inserted %d at position %d.\n", value, pos);
89     }
90 }
91 int main() {
92     int choice, n, value, pos;
93
94     while (1) {
95         printf("\n--- Singly Linked List Menu ---\n");
96         printf("1. Create a List\n");
97         printf("2. Display the List\n");
98         printf("3. Insert at Beginning\n");
99         printf("4. Insert at End\n");
```

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```
90 }
91 int main() {
92     int choice, n, value, pos;
93
94     while (1) {
95         printf("\n--- Singly Linked List Menu ---\n");
96         printf("1. Create a List\n");
97         printf("2. Display the List\n");
98         printf("3. Insert at Beginning\n");
99         printf("4. Insert at End\n");
100        printf("5. Insert at Position\n");
101        printf("6. Exit\n");
102        printf("Enter your choice: ");
103        scanf("%d", &choice);
104
105        switch (choice) {
106            case 1:
107                printf("Enter number of nodes: ");
108                scanf("%d", &n);
109                createList(n);
110                break;
111            case 2:
112                displayList();
113                break;
114            case 3:
115                printf("Enter value to insert: ");
116                scanf("%d", &value);
117                insertAtBeginning(value);
118                break;
119            case 4:
120                printf("Enter value to insert: ");
121                scanf("%d", &value);
122                insertAtEnd(value);
123                break;
124            case 5:
125                printf("Enter position: ");
126                scanf("%d", &pos);
127                printf("Enter value: ");
128                scanf("%d", &value);
129                insertAtPosition(value, pos);
130                break;
131            case 6:
132                exit(0);
133            default:
134                printf("Invalid choice!\n");
135        }
136    }
137
138    return 0;
139 }
140 }
```

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"C:\Users\Admin\Desktop\18M24CS173\Single linked list.exe"

--- Singly Linked List Menu ---

1. Create a List
2. Display the List
3. Insert at Beginning
4. Insert at End
5. Insert at Position
6. Exit

Enter your choice: 4
Enter value to insert: 1
Inserted 1 at the end.

--- Singly Linked List Menu ---

1. Create a List
2. Display the List
3. Insert at Beginning
4. Insert at End
5. Insert at Position
6. Exit

Enter your choice: 2
Linked List: 34 -> 23 -> 18 -> 45 -> 9 -> 1 -> NULL

--- Singly Linked List Menu ---

1. Create a List
2. Display the List
3. Insert at Beginning
4. Insert at End
5. Insert at Position
6. Exit

Enter your choice: 5
Enter position: 3
Enter value: 44
Inserted 44 at position 3.

--- Singly Linked List Menu ---

1. Create a List
2. Display the List
3. Insert at Beginning
4. Insert at End
5. Insert at Position
6. Exit

Enter your choice: 2
Linked List: 34 -> 23 -> 44 -> 18 -> 45 -> 9 -> 1 -> NULL

--- Singly Linked List Menu ---

1. Create a List
2. Display the List
3. Insert at Beginning
4. Insert at End
5. Insert at Position
6. Exit

Enter your choice: 6

Process returned 0 (0x0) execution time : 97.239 s
Press any key to continue.

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